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VW 113 - To what extent is the 30-day time window of the Liquidity Coverage Ratio (LCR) adequate under the conditions of a digital bank run, and how long could UBS Group, based on its current liquidity position, withstand an outflow scenario analogous to the 2023 Credit Suisse crisis?

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Abstract

This paper examines how long UBS Group would withstand a digital *bank run* that follows the outflow pattern of the Credit Suisse crisis of March 2023. The starting point is the theoretical insight that banks, owing to their maturity transformation, are structurally vulnerable to coordinated deposit withdrawals and that digitalisation has massively accelerated this dynamic. As a regulatory response, the Liquidity Coverage Ratio (LCR) under Basel III prescribes a 30-day liquidity buffer. Using Credit Suisse's (CS) daily outflow data, the paper shows that in the crisis this window was undershot many times over: the average daily outflow rate was 2.24 times the LCR standard. A short-term stress test of UBS Group built on this shows that its High-Quality Liquid Assets (HQLA) base would be exhausted after about 14 days in the CS benchmark scenario and after only 5 days in the extreme scenario of a digital *flash run*. These results indicate that the LCR's 30-day window is structurally miscalibrated for digital crisis scenarios and that a globally systemically important bank cannot survive such a run on its own.

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1. Introduction

The collapse of Credit Suisse (CS) in March 2023 marks a turning point in the history of modern financial crises. Within five trading days, the bank lost CHF 44.7 bn of liquidity, equivalent to around 37% of its total HQLA holdings (Eidgenössische Finanzmarktaufsicht [FINMA], 2023, pp. 37–38). The speed and intensity of these outflows exceeded the assumptions of the regulatory standard model, the Liquidity Coverage Ratio (LCR) under Basel III, many times over – the average daily outflow rate was 2.24 times the LCR standard (cf. Ch. 3.2). The Basel Committee had calibrated the LCR on the basis of the crisis patterns of 2007–2009 and defined a 30-day survival horizon as the standard (Basel Committee on Banking Supervision [BCBS], 2017, p. 1). The Credit Suisse crisis shows that the digitalisation of banking fundamentally undermines this time assumption.

Against this background, this paper examines the following research question: To what extent is the 30-day time window of the Liquidity Coverage Ratio (LCR) adequate under the conditions of a digital *bank run*, and how long could UBS Group, based on its current liquidity position, withstand an outflow scenario analogous to the 2023 Credit Suisse crisis?

A three-stage approach is chosen to answer this question. First, Chapter 2 lays the theoretical foundations: it shows how maturity transformation makes banks structurally vulnerable to runs (Ch. 2.1), how the LCR addresses this risk from a regulatory perspective (Ch. 2.2), and why digitalisation erodes the time assumptions underlying the model (Ch. 2.3). Chapter 3 translates these theoretical arguments into an empirical diagnosis by reconstructing CS's daily outflow rates and identifying the concrete weaknesses of the LCR. On this basis, Chapter 4 develops a deterministic short-term stress test that applies the CS outflow rates as a *benchmark* to UBS Group's current liquidity base and calculates survival time under three scenarios. Chapter 4.3.2 places the results in a regulatory context.

The paper thus contributes to the current debate on the adequacy of the LCR calibration in the age of digital bank runs – a debate that the Swiss

Federal Council, in its Too-Big-to-Fail report (TBTF report) of April 2024, has classified as a regulatory priority (Eidgenössisches Finanzdepartement [EFD], 2024, p. 38).

2. Theoretical Foundations: Liquidity and Bank Runs

In order to assess the resilience of banks in modern crisis scenarios, a well-founded understanding of their structural functioning and of the resulting regulatory framework is needed first. This chapter lays the theoretical and conceptual foundations for the later empirical analysis. To this end, Section 2.1 first examines the role of banks as intermediaries. It explains how the economically necessary maturity transformation makes banks fundamentally vulnerable to liquidity crises. As a regulatory response to this inherent fragility, Section 2.2 presents the Liquidity Coverage Ratio (LCR) under Basel III. Building on this, Section 2.3 analyses the technological paradigm shift of the digital bank run and shows how the speed of information and transactions, massively accelerated by digitalisation and social media, challenges the static premises of the LCR.

2.1. Banks as Intermediaries between Short-Term Deposits and Long-Term Loans

In the modern financial system, banks assume a central role as intermediaries by transforming short-term, on-demand deposits into long-term, illiquid loans (Diamond, 2007, p. 189). This maturity and liquidity transformation is economically significant because it enables investments with long commitment periods and thus contributes to overall economic development (Diamond, 2007, pp. 190–191). At the same time, however, it creates a fragility, since the liability side (short-term deposits) of the bank is substantially more liquid than its asset side (banks' long-term lending). Diamond and Dybvig (1983, pp. 401–419) describe how banks thereby become vulnerable as soon as many depositors demand liquidity at the same time and force an early, value-destroying liquidation of assets. In such situations, banks get into difficulty not because of a lack of solvency but because of illiquidity (Diamond & Dybvig, 1983, p. 410).

The theoretical basis is developed in the model by Diamond and Dybvig (1983). The central insight is that at time $T = 0$ depositors do not know whether they will need liquidity early ($T = 1$) or late ($T = 2$). This individual liquidity risk

is formalised via two types of consumers: "early" consumers (type 1), who consume exclusively at $t = 1$, and "late" consumers (type 2), whose consumption takes place only at $t = 2$. Each depositor becomes type 1 with probability t; the remaining mass $1 - t$ becomes type 2. The bank's investment opportunities are illiquid: one unit of the initial endowment (*endowment*) invested at $T = 0$ can, in the event of early termination at $T = 1$, be liquidated only at its investment value of 1, but yields a higher return of $R > 1$ if held to full maturity until $T = 2$ (Diamond & Dybvig, 1983, p. 406). The gross return $R > 1$ means that for every unit invested at $T = 0$ the bank receives the amount R at $T = 2$; the implicit interest rate is thus $R-1$.

Since type membership is private information, competitive markets cannot offer efficient insurance against liquidity shocks. The bank, by contrast, implements collective insurance through demand deposits: it offers early consumers a payout c_1 in $T = 1$ and late consumers a payout c_2 in $T = 2$. Under full information, the bank maximises the expected utility

subject to the constraint

$$tc_1 + \frac{(1-t)c_2}{R} = 1.$$

This budget constraint states that the bank must divide its total resources (1 unit) such that it can pay out the early consumers (share t) immediately and leave the resources for the late consumers (share $1 - t$) invested until $T = 2$, the latter earning the higher return R. From the first-order conditions

$$u'(c_1) = Ru'(c_2)$$

it follows that in the optimum $1 < c_1 < c_2 < R$ holds (Diamond, 2007, p. 192). With a concave utility function ($u'' < 0$), the first-order condition implies that $c_1 < c_2$ must hold, since $R > 1$. At the same time $c_1 > 1$, because the bank offers early consumers more than the liquidation option, and $c_2 < R$, because part of the resources has already flowed out at $T = 1$. The bank creates liquidity by enabling early consumers a payout $c_1 > 1$ – that is, more than they would receive from independently liquidating the investment early (only 1 there). This added value is the economic contribution of the bank as an intermediary.

This insurance effect, however, is what gives rise to vulnerability to bank runs. A decisive feature of the deposit contract is the *sequential service constraint* (Diamond & Dybvig, 1983, p. 408): the bank pays out depositors in the order in which they appear, without knowing the total number of withdrawals in advance. A bank run arises as an alternative Nash equilibrium[1] as soon as late consumers anticipate that so many depositors will withdraw that the future payout value V_2 falls below the current payout value V_1. At that moment, it becomes rational for a type-2 consumer to withdraw immediately as well (Diamond & Dybvig, 1983, p. 409). The fragility of the bank is a direct consequence of its economic function: precisely because it creates added value for depositors through maturity transformation ($c_1 > 1$), the possibility of a bank run as an alternative equilibrium arises.

The Diamond-Dybvig model forms the theoretical core of the bank-run literature and has been taken up in numerous extensions, including the integration of interbank markets and the analysis of efficient crises (Allen & Gale, 1998), the unique determination of run probabilities by means of *global games* (Goldstein & Pauzner, 2005), the study of time-inconsistent government intervention promises (Ennis & Keister, 2009), and the conceptual distinction between panic-based and information-based runs (Bhattacharya & Jacklin, 1988). However, the model also has limitations. Gorton (1988, p. 752) emphasises that historical banking panics were systematic reactions of depositors to changed risk perceptions and were not based on random signals, so-called *sunspots*[2] (Diamond & Dybvig, 1983, p. 410). Goldstein and Pauzner (2005, pp. 1294–1307) develop this objection further and show in a global-games model that runs do not arise exclusively as random coordination failures but can also be triggered by fundamental information about bank quality. This distinction is of direct relevance for the present paper: the run on CS in March 2023 was not a pure sunspot event but an information-driven reaction to fundamental weaknesses of the bank, accelerated by digital coordination mechanisms (cf. Ch. 2.3). Moreover, the Diamond-Dybvig model neglects the

1 A Nash equilibrium is a situation in which no individual actor (here: depositor) can gain by deviating from his or her strategy as long as all others keep their decisions unchanged (Kolmar, 2021, p. 276).

2 In the bank-run literature, sunspots denote purely extrinsic signals that have no connection to the economic fundamentals of a bank but can lead to panic reactions if depositors believe that others will also react to them (Gu, 2011, pp. 398–400). In the context of the model, this means that consumers react to noise or random, irrelevant information, for example media reports or socially shared rumours.

dynamics of modern financial markets: the assumption of a physical queue is replaced in the digital age by real-time transactions, which massively accelerates the coordination speed of depositors (FINMA, 2023, p. 67).

2.2. Concepts of Liquidity Measurement: The Liquidity Coverage Ratio (LCR) under Basel III

The preceding chapter has shown that banks are structurally fragile: because their liabilities are considerably more liquid than their assets, the mere expectation of a run can trigger the sequential service constraint and drive a bank into illiquidity despite solvency. Regulation must solve this coordination problem ex ante, before a run arises.

The most important regulatory response is the Liquidity Coverage Ratio (LCR), introduced by the Basel Committee on Banking Supervision (BCBS) in response to the financial crisis of 2007–2009 (BCBS, 2017, p. 1). Its objective is concrete: in a severe but plausible stress scenario, a bank should survive 30 calendar days on its own, without relying on emergency central bank loans (BCBS, 2019, p. 3 [LCR30]). To this end, it must hold a buffer of first-class, unencumbered liquid assets (*High-Quality Liquid Assets*, HQLA) that covers at least the expected net cash outflows of this scenario (BCBS, 2019, pp. 3–4 [LCR20]). Formally, the ratio reads:

$$LCR = \frac{\textit{Stock of HQLA}}{\textit{Net liquidity outflows over the next 30 days}} \geq 100\%$$

The following analysis is organised along this formula: first the numerator (HQLA) is explained, then the denominator (outflow rates) is discussed, and finally it is shown why the 30-day window represents a central weakness.

2.2.1. Numerator of the LCR: High-Quality Liquid Assets (HQLA)

HQLA are assets that can be converted into cash quickly and without a large loss of value in periods of stress (BCBS, 2019, p. 4 [LCR30]). The rules require that they be unencumbered, eligible for central bank operations, and tradable in deep, liquid markets (BCBS, 2019, pp. 4–5 [LCR30]). Level 1 assets comprise assets of the highest quality: cash, central bank reserves, and government bonds with a 0% risk weight, for example US Treasury bonds or German federal bonds (Verordnung über die Liquidität der Banken und Wertpapierhäuser [LiqV], 2012, Art. 15a; BCBS, 2019, pp. 14–15 [LCR30]). They can be

counted in full and without a valuation haircut (*haircut*) (BCBS, 2019, p. 14 [LCR30]; LiqV, 2012, Art. 15a para. 3). Level 2 assets have lower liquidity and are subject to haircuts and a cap of at most 40% of total HQLA holdings (BCBS, 2019, p. 16 [LCR30]). Level 2a assets are subject to a haircut of 15% (BCBS, 2019, pp. 16–17 [LCR30]), Level 2b assets under Swiss law to at least 50% (LiqV, 2012, Art. 15b para. 4, Art. 15b para. 6; BCBS, 2019, pp. 19–22 [LCR30]).

2.2.2. Denominator of the LCR: Net Liquidity Outflows and Run-off Rates

The denominator of the LCR results from the expected outflows less the expected inflows over 30 days, whereby inflows are capped at 75% of outflows (BCBS, 2019, p. 3 [LCR40]). Standardised *run-off* rates are assigned to the outflows depending on their category. Run-off rates denote the outflow ratios assumed by regulation over the 30-day horizon. A run-off rate of 5% means that 5% of the respective deposit category flows out within 30 days. Retail deposits are considered relatively inert (Argyle et al., 2026, pp. 1–6): stable deposits with deposit insurance receive a run-off rate of 5%, less stable deposits rates of 10% or more (LiqV, 2012, Annex 2 in conjunction with Art. 16 para. 3). Unsecured wholesale funding is treated with higher rates. Operational deposits receive moderate rates of 25–40% (BCBS, 2019, pp. 12–15 [LCR40]; LiqV, 2012, Annex 2 in conjunction with Art. 16 para. 3), while non-operational deposits from financial institutions are charged up to 100% (BCBS, 2019, pp. 2–3 [LCR99]; LiqV, 2012, Annex 2 in conjunction with Art. 16 para. 3). This maximum rate reflects the insight that professional market participants withdraw immediately at the first doubts. Secured funding is differentiated according to the quality of the collateral. In simplified terms: if a bank holds CHF 100 bn in stable retail deposits (run-off rate 5%) and CHF 50 bn in non-operational wholesale deposits (run-off rate 100%), the expected 30-day outflow is CHF 55 bn (5 + 50). Gorton and Metrick (2012, pp. 425–428) show that modern crises often arise primarily through such a *run on repo*: as soon as institutional lenders doubt the value of the collateral, they refuse to roll over the funding.

2.2.3. Rationale and Limits of the LCR

The LCR addresses the coordination problem directly: by prescribing minimum liquidity, it is meant to prevent a bank from being forced into value-destroying emergency liquidation upon coordinated withdrawals (cf. Ch. 2.1). The decisive premise of the

model, however, is the 30-day window. It assumes that outflows occur linearly and predictably spread over one month. Persaud (2008, pp. 75–78) already criticised the procyclical effect of static liquidity ratios, since their assumptions fail precisely at the moment when they are needed most urgently. Brunnermeier and Pedersen (2009, pp. 2202–2204) also show that market liquidity and funding liquidity can fall into a mutually reinforcing downward spiral in crises, which further calls into question the linear time assumption of the LCR. The European Central Bank confirms this assessment and explicitly states that the LCR is not designed to cover all extreme events involving deposit outflows (Europäische Zentralbank [EZB], 2023).

This weakness is not limited to the LCR alone. The *Net Stable Funding Ratio* (NSFR), designed as its long-term counterpart, requires that available stable funding exceed required stable funding over a one-year horizon (BCBS, 2014, p. 2). But the NSFR, too, operates on aggregated time periods and captures neither *intraday* dynamics nor the nonlinear acceleration of outflows in a crisis (BCBS, 2014, p. 3).

The entire Basel III liquidity regime thus rests on a time assumption of gradual outflows that is structurally undermined by digitalisation. As the following Chapter 2.3 will show, the technological reality of digital banking erodes precisely this assumption.

2.3. The Digital Bank Run: Theory of Accelerated Information Diffusion and Transaction Speed

The LCR models liquidity outflows over a horizon of 30 calendar days. This window is a regulatory design decision of the Basel Committee: it is meant to give supervisors and bank management sufficient time to take corrective measures in a stress situation. The model thus implicitly assumes that outflows are spread over this period, an assumption based on the crisis patterns of the financial crisis of 2007–2009. Technological change over recent decades has fundamentally altered this dynamic. Three key mechanisms accelerate information diffusion and transaction speed to such an extent that the 30-day window appears structurally insufficient in the case of a digital bank run (Ofir & Elmakiess, 2025, pp. 15–16; Bindseil & Senner, 2023, p. 33).

The first driver is the technological time effect. The widespread implementation of *24/7 fast payment systems* makes it possible to move funds around the clock and regardless of location (The Federal Reserve, 2023, p. 53). Deposits thereby turn into so-called *hot money*: funds that can be withdrawn immediately at the slightest sign of instability (Benmelech et al., 2023, p. 1; Ofir & Elmakiess, 2025, p. 16). Historical comparisons of the banking crises of 1984, 2008, and 2023 show that the speed and volume of deposit outflows have increased drastically as a result of these innovations (Rose, 2023; The Federal Reserve, 2023, p. 35; Krogstrup et al., 2024, p. 48).

The second factor is the coordination effect, which acts directly on the game-theoretic structure of the Diamond-Dybvig model (cf. Ch. 2.1). In the original model, a bank run arises because depositors cannot observe what others are doing; the inefficient equilibrium is thus an information problem. Social networks solve this problem, albeit in a destabilising direction: they enable depositors to communicate in real time and to coordinate collective action (Bindseil & Senner, 2023, p. 36). Misinformation and negative sentiment spread virally and act as a digital sunspot (Cookson et al., 2023, Abstract; Fofanov, 2025, pp. 53–54). Empirical studies on the collapse of Silicon Valley Bank in March 2023 show that an increased volume of negative social media posts correlates with immediate outflows of uninsured deposits and hourly price slumps (The Federal Reserve, 2023, p. 35). In light of the global-games approach of Goldstein and Pauzner (2005, pp. 1294–1307), this finding can be classified in a more nuanced way: social media reduces the uncertainty about the behaviour of other depositors and thereby shifts the threshold at which a run becomes rational considerably downwards (i.e. the critical signal threshold at which a depositor withdraws falls).

The third mechanism is the psychological effect, which is based on the reduction of decision frictions and the emergence of information cascades. Bikhchandani et al. (1992, pp. 992–1000) formalise this process: as soon as a critical mass of actors chooses a particular action, it becomes rational for subsequent actors to follow this decision regardless of their own information. Private information is dominated by the observed action of others, which leads to a fragile uniformity. Shiller (2015, pp. 175–187) transfers this logic to financial markets and shows that herd behaviour is amplified by emotional contagion in periods of stress, with digital media acting as a multiplier. In the context of a bank run, this means: the impersonal, immediate nature of digital transactions removes the psychological barrier of a

physical branch visit (Ofir & Elmakiess, 2025, pp. 21–23), while the visibility of collective withdrawals via social media lowers the threshold for an information cascade. Both effects favour the realisation of the inefficient run equilibrium.

In summary, digitalisation has fundamentally transformed the architecture of a bank run: the time effect, coordination via social media, and the removal of psychological frictions compress outflow dynamics, which the LCR model spreads over 30 days, into a few hours or days. Against this theoretical background, the following chapter reconstructs the actual outflow rates of CS in March 2023 in order to use them as an empirical benchmark for the UBS analysis in Chapter 4.

3. Analysis of the Credit Suisse Scenario as a Benchmark

The theoretical foundation in Chapter 2 has developed three theses: first, banks are structurally vulnerable to runs because of maturity transformation (cf. Ch. 2.1). Second, the LCR is meant to mitigate this risk ex ante through a 30-day liquidity buffer (cf. Ch. 2.2). Third, digitalisation erodes precisely the time assumption on which this buffer rests (cf. Ch. 2.3). This chapter translates these theoretical arguments into an empirical diagnosis: the collapse of CS in March 2023 serves as a benchmark because it is the most recent and most thoroughly documented case of a digital bank run at a globally systemically important bank, whose daily outflow data are officially documented by the FINMA report (2023, pp. 37–38).

3.1. Reconstruction of CS Outflow Rates in March 2023

The collapse of CS did not happen abruptly but was the result of a months-long erosion of confidence that culminated in an acute digital bank run in March 2023 (FINMA, 2023, pp. 6–7). Already in the fourth quarter of 2022, CS lost CHF 111 bn in deposits as a result of a series of scandals, management changes, and share price losses (Martins, 2024, p. 1126; Richter, 2023). The LCR thereby fell from 203% at the end of 2021 to 144% at the end of 2022, and the available liquidity buffer shrank to around CHF 100 bn (FINMA, 2023, p. 63; Credit Suisse Group, 2023, p. 7; p. 33). This weakened starting position is central to the later diagnosis: CS entered the acute crisis with an already considerably depleted buffer.

The immediate trigger was a double shock. On 10 March, Silicon Valley Bank collapsed in the US (FINMA, 2023, p. 68), an event without direct relevance for CS's fundamentals, but which, in the sense of Goldstein and Pauzner (2005), acted as an informational trigger: it abruptly changed the risk perception of institutional depositors worldwide (cf. Ch. 2.1). Two days later, CS's main shareholder, the Saudi National Bank, publicly declared that it did not wish to invest any further funds (FINMA, 2023, p. 68; Böni & Zimmermann, 2024, p. 6). This statement spread within hours via news channels and social media and triggered the collective withdrawal of institutional depositors (EFD, 2024, p. 58; The Federal Reserve, 2023, p. 53; Cookson et al., 2023, Abstract).

The daily outflow data of the FINMA report (2023, pp. 37–38) document the acceleration of this process.

Table 1

Outflow data of Credit Suisse during the bank run in March 2023

Day	**Date**	**Outflow (CHF bn)**	**% of HQLA**
Monday	13 March 2023	-1.6	-1.33 %
Tuesday	14 March 2023	-2.7	-2.25 %
Wednesday	15 March 2023	-13.2	-11.0 %
Thursday	16 March 2023	-17.1	-14.25 %
Friday	17 March 2023	-10.1	-8.42 %
Total	**5 trading days**	**-44.7**	**-37.26 %**

Note: Own illustration based on FINMA (2023, pp. 37–38). Created using Figma.

Within five trading days, CS lost CHF 44.7 bn; around two thirds of the total quarterly outflows of CHF 67 bn were thus concentrated in less than a week (Richter, 2023; FINMA, 2023, p. 36; Credit Suisse Group, 2023, p. 7). On 16 March, the Federal Council created, by way of an ordinance, the legal basis for extraordinary liquidity assistance from the SNB (FINMA, 2023, p. 38). On 19 March, the emergency ordinance was adopted, which enabled the merger with UBS Group (FINMA, 2023, p. 40).

The composition of the outflows confirms the theoretical pattern from Chapter 2.2: institutional and wealthy depositors withdrew first and most heavily. The *wealth management* segment lost around 9% of its assets under management in the first quarter of 2023 (Credit Suisse Group, 2023, p. 7). Retail deposits from the Swiss core business, by contrast, initially remained more stable, which corresponds to the model logic of low run-off rates for stable retail client deposits (Credit Suisse Group, 2023, p. 7). This pattern is consistent with the theory: institutional actors have the information and infrastructure to react immediately, while private depositors are slowed down by deposit insurance and higher switching costs.

3.2. Identification of Weaknesses in the Standard LCR Model

The outflow data now allow a precise diagnosis of which specific assumptions of the LCR model did not hold in the CS case. The following analysis focuses on two points of failure that are empirically observable in the CS case and that serve as direct input parameters for the stress test in Chapter 4: the time window and the calibration of the run-off rates.

The LCR model assumes that stress outflows are spread over 30 calendar days and that the bank can deploy its HQLA buffer gradually (cf. Ch. 2.2). The CS case refutes this empirically. Although the rules do not prescribe daily outflow rates but sum liquidity holdings ex ante over a 30-day horizon, a computational linearisation allows a direct comparison with reality: under the simplifying assumption of an even distribution, a daily rate of:

$$\textit{Daily rate} = \frac{100\%}{30\ \textit{days}} \approx 3.3\%\ \textit{per day}$$

results. On 15 March alone, 13.2% of the liquidity base flowed out, which corresponds to four times the LCR daily rate (FINMA, 2023, p. 38). On 16 March it was 17.1%, more than five times (FINMA, 2023, p. 38). Over the entire critical week, the

average daily rate was 7.45%, i.e. 2.24 times that of the LCR model. The bank was thus not illiquid because its HQLA buffer was absolutely too small; it was illiquid because the outflows exceeded the model many times over in the time dimension (FINMA, 2023, p. 7).

The second structural problem lies in the temporal distribution of run-off rates for institutional deposits (FINMA, 2023, p. 72). The LCR model does apply rates of up to 100% to non-operational deposits from financial institutions, but distributes them evenly over 30 days. The CS case shows the opposite: institutional actors did not react gradually but discretely and collectively as soon as the coordination mechanism was triggered via social media and market channels (FINMA, 2023, p. 7). This finding confirms the theoretical predictions of Brunnermeier and Pedersen (2009, pp. 2203–2204) on the liquidity spiral: as soon as an outflow exceeds a critical threshold, it reinforces itself, since falling asset prices trigger further *margin calls*. Glossner et al. (2022, pp. 19–20) additionally argue that institutional actors must withdraw their funds quickly in crisis phases because of their own liquidity pressure and not at the daily rates assumed by the model. The run thus did not proceed as the LCR model simulates it, but as the Diamond-Dybvig model describes it theoretically: as a collective equilibrium event, not as a linear process (cf. Ch. 2).

From the empirically observed average of 7.45% per day arises the central question of this paper: how long would UBS Group, with its current liquidity base, withstand an equivalent scenario? Chapter 4 develops for this purpose a short-term stress test that uses the CS daily rates as benchmark parameters in order to answer the research question formulated in the introduction.

4. Empirical Stress Test: The UBS Group in the 5-Day Scenario

The preceding chapter has shown that the actual outflow rates of CS massively exceeded the LCR model in the time dimension. The following chapter now examines whether UBS Group could withstand a comparable scenario. To this end, the data basis is presented first (Ch. 4.1), then the methodology of the stress test is developed (Ch. 4.2), before the results are presented and discussed (Ch. 4.3).

4.1. Data Basis: Analysis of HQLA and Deposit Structure

The basis of the stress test is UBS Group's official LCR data from the Pillar 3 Report 2025 (pp. 89–90). Since UBS Group, as a global systemically important bank (G-SIB), is subject to consolidated reporting, all positions are weighted according to the stricter Swiss requirements under the LiqV and FINMA. All values are year-end values as of 31 December 2025 in USD million.

On the asset side, UBS Group has total HQLA of USD 331'568 million, of which USD 302'112 million in Level 1 assets (cash, central bank reserves, and government bonds with a 0% risk weight, no haircut) and USD 29'456 million in Level 2 assets (already after deduction of regulatory haircuts). The high share of Level 1 assets of 91% reflects the conservative liquidity strategy of post-CS-takeover UBS.

On the liability side, the weighted gross outflows over 30 days amount to USD 390'134 million. The reported gross outflows are based on the regulatory LCR stress scenario under Basel III and are taken from UBS's official Pillar 3 data. The largest outflow components are non-operational wholesale deposits (USD 123'894 million), secured funding (USD 103'944 million), and other positions from derivatives and credit lines (USD 45'780 million). The weighted inflows amount to USD 208'441 million and are below the 75% cap, so they are fully eligible. This results in net outflows of USD 181'693 million and an LCR of around 182.5% (year-end value). Table 2 summarises the key figures.

Table 2

Liquidity positions of UBS (as of 2025)

Position	Value (in USD million)
Level 1 HQLA	302'112
Level 2 HQLA	29'456
Total HQLA	**331'568**
Gross outflows	390'134
Inflows	208'441
Net outflows	**181'693**
LCR	**182.5 %**

Note: Own illustration based on UBS Group (2025, pp. 89–90). Created using Figma & Python.

An additional limitation that the stress test, by construction, does not capture concerns the operational availability of HQLA. In the aftermath of the CS crisis, the BCBS notes that operational liquidity needs, in particular intraday payment flows and counterparty *prepositioning*, absorbed almost all of CS's LCR net outflows (BCBS, 2024, Section 1). The HQLA holdings of UBS Group reported here should therefore be interpreted as gross holdings; the liquidity reserve actually freely available could be lower. The BCBS does not quantify the share exactly but notes that operational needs at CS ‹almost entirely absorbed the total net cash outflows› (BCBS, 2024, Section 1), which suggests that the effective reserve could be significantly below the reported HQLA holdings.

4.2. Methodology: Development of a Short-Term Stress Test

The stress test follows a linear point-estimate model (*point estimate*). The choice of a deterministic approach instead of a Monte Carlo simulation is deliberate: whereas stochastic models make the results heavily dependent on the assumed distributions, the chosen approach allows full transparency of the causal structure (Sorge, 2004, p. 8; Borio et al., 2012, pp. 2–7; Kupiec, 2018, pp. 144–145). The daily net outflow is constant, and the survival time results directly as the quotient of the

available HQLA base and the daily *drain*. Every assumption and its influence on the result can thus be read off directly.

The model contains one central stress parameter: the scaling factor k, which raises the daily gross outflow rate relative to the Basel III standard. The daily net outflow under a scenario (k, f) results formally as:

$$\textit{Daily net outflow} = \left(\frac{\textit{Gross outflows}_{30t}}{30}\right) \cdot k - \left(\frac{\textit{Inflows}_{30t}}{30}\right) \cdot f$$

where f denotes the share of inflows that still arrive under stress and k the intensity multiplier on the outflow side. The approach of scaling existing outflow rates by an empirically derived factor k is established in the stress-test literature: the Basel Committee itself calibrated the LCR run-off rates through scaled observations from the financial crisis of 2007–2009; the factor $k = 2.24$ determined empirically here shows that the CS crisis clearly exceeded the calibration basis (BCBS, 2017, p. 1).

On this methodological basis, three scenarios are derived:

Scenario A serves as the reference scenario and corresponds to the regulatory standard LCR model without modification (k = 1.0). This allows a direct check of whether UBS Group meets the 30-day survival criterion.

Scenario B represents the CS benchmark $(k = 2.24)$. The actual average daily outflow rate of CS in the crisis week was 7.45% of the liquidity base, compared with an LCR-implied daily rate of 3.3%. The scaling factor thus results as k = 7.45/3.3 ≈ 2.24 (FINMA, 2023, pp. 37–38). With regard to inflows, Cipriani et al. (2024, pp. 14–15) show that run banks received about 20–28% fewer payments from counterparties in March 2023. The inflow factor of $f = 0.72$ thus represents a slightly conservative but empirically grounded assumption.

Scenario C models a digital *flash run* $(k = e^{1.54} \approx 4.66)$. For this most extreme scenario, the empirical finding of Cipriani et al. (2024, p. 14) is used: on the stress days of 10 and 13 March 2023, the total payment value of run banks was increased by a factor of $e^{1.54} \approx 4.66$, statistically significant at the 1% level. Since Level 2 assets can no longer be realised within a few hours in a panic environment, only the Level 1 base (USD 302'112 million) is applied as available HQLA for Scenario C (BCBS,

2013, paragraph 25). Inflows are set to 0 $(f = 0)$, since the largest banks drastically reduced their payments to run banks in March 2023 (Cipriani et al., 2024, p. 14). The scenario parameters are summarised in Table 3.

The linearity assumption represents the most important methodological limitation of the model. In reality, runs typically accelerate: CS outflows rose from 1.6% on Monday to 17.1% on Thursday (cf. Ch. 3.1). Brunnermeier and Pedersen (2009, pp. 2203–2204) formalise this mechanism as a liquidity spiral in which falling market liquidity leads to rising haircuts and thus to further fire sales (cf. Ch. 2.2.3). A linear model thus underestimates the speed of early depletion and overestimates the actual time available for management to act. The survival times estimated below should therefore be interpreted as upper bounds; the actual survival time could be shorter. A convex outflow profile, as the CS data suggest (cf. Tab. 1), would bring forward the depletion, since the bulk of the outflows would be concentrated in the first days.

Besides the linearity assumption, the model has two further limitations that must be taken into account when interpreting the results. The transfer of CS outflow rates to UBS Group rests on the assumption that both banks have a comparable depositor structure and outflow characteristics. Greenlaw et al. (2012, pp. 8–14), however, show that stress tests that treat banks in isolation and with uniform stress parameters can systematically misjudge the actual vulnerability of individual institutions, since the distribution of losses depends substantially on the bank-specific portfolio and depositor structure. Although UBS Group and CS, as G-SIBs headquartered in Switzerland, were both subject to comparable regulatory frameworks, their business models differ, which limits the transferability of the outflow intensity. The scenarios used here are therefore to be understood as illustrative benchmarks, not as bank-specific forecasts.

Furthermore, the model cannot capture feedback effects between outflows, asset prices, and counterparty behaviour. Greenwood et al. (2014, pp. 472–475) show that a bank's fire sales can trigger cascading losses at other institutions via common portfolio exposures, a mechanism that, by construction, is not captured in a single-bank stress test. The model thus represents the bank's first, isolated line of defence without taking systemic amplification effects into account. The

reported survival times are therefore to be interpreted as upper bounds in three respects: the linear model underestimates the outflow speed, the transfer of uniform parameters possibly overstates comparability, and the omission of feedback effects underestimates systemic stress.

4.3. Results and Discussion

Table 3 shows the results of the three scenarios.

Table 3

Scenario analysis of the short-term stress test

Scenario	Intensity multiplier k	Inflow factor f	HQLA base (USD million)	Daily net outflow (USD million)	Survival time (days)
A - Basel 3 Standard	1.00	100 %	331'568	6'056	54.7
B - CS-Benchmark	2.24	72 %	331'568	24'127	13.7
C - Digital Flash Run	4.66	0 %	302'113	60'661	5.0

Note: Own illustration. Created using Figma & Python.

Scenario A confirms that UBS Group comfortably meets the regulatory 30-day survival criterion of the LCR with an LCR of approx. 182.5%. Even with a complete loss of inflows, the HQLA base would be exhausted only after approx. 25 days.

Scenario B shows that with an outflow pattern analogous to the CS crisis, UBS Group's HQLA would be completely used up after around 14 days. The 30-day window of the LCR would thus not even last half as long. UBS Group is indeed considerably better capitalised than CS at the beginning of 2023 (LCR 182.5% vs. 144%), but its daily drain in this scenario reflects an absolutely larger balance sheet with a similar outflow intensity. This finding confirms the theoretical prediction of Acharya and Mora (2015, p. 3), according to which larger banks are not proportionally more resilient in a crisis, since their outflow volumes scale with balance sheet size.

Scenario C shows the most critical picture: with outflows at the level of the peak days of the March 2023 run, UBS Group, using only its Level 1 assets, is left with just 5 days. This corresponds roughly to the duration after which CS depended on state intervention in March 2023 (FINMA, 2023, pp. 37-39).

4.3.1. Sensitivity Analysis

To test the robustness of the results against parameter uncertainty, Table 4 systematically varies the k factor from 1.0 to 5.0 under three inflow assumptions.

Table 4

Liquidity survival time as a function of intensity multipliers and inflow scenarios

Intensity multiplier k	HQLA base	Full inflows (100 %)	CS inflows (72 %)	No inflows (0 %)
1.00	L1 + L2	54.7	41.4	25.5
1.10	L1 + L2	45.1	35.6	23.2
1.20	L1 + L2	38.3	31.3	21.2
1.30	L1 + L2	33.3	27.9	19.6
1.40	L1 + L2	29.5	25.1	18.2
1.50	L1 + L2	26.4	22.9	17.0
2.00	L1 + L2	17.4	15.8	12.7
2.24 (CS comparison)	L1 + L2	14.9	13.7	11.4
2.50	L1 + L2	13.0	12.1	10.2
3.00	L1 + L2	10.3	9.7	8.5
3.50	L1 + L2	8.6	8.2	7.3
4.0	L1 + L2	7.4	7.1	6.4
4.66 (flash run comparison)	L1	5.6	5.4	5.0
5.0	L1	5.2	5.0	4.6

Note: Own illustration. Created using Figma & Python.

The sensitivity analysis yields two insights. First, the k factor is the dominant driver of survival time: a doubling of k roughly halves the survival time. The inflow factor f has a considerably smaller influence by comparison, since inflows make up only around 35% of outflows. Second, the table shows a critical threshold at $k \approx 1.4$: from here on, UBS Group leaves the 30-day survival safety zone under normal inflows ($f = 100\ \%$). Outflow rates that are only 40% above the Basel III standard

thus already endanger the regulatory objective. This underscores the thesis developed in Chapter 2.3 that the LCR model has a structural weakness in its time dimension.

An obvious counter-question is whether survival time could be secured by a higher HQLA cushion. Table 5 answers this question by systematically scaling the current HQLA holdings.

Table 5

HQLA sensitivity analysis of the short-term stress tests

Increase in HQLA	HQLA absolute (USD million)	Scenario A (days survived)	Scenario B (days survived)	Scenario C (days survived)
0 %	331'568	54.7	13.7	5.0
5 %	348'146	57.5	14.4	5.2
10 %	364'146	60.2	15.1	5.5
20 %	397'882	65.7	16.5	6.0
50 %	497'352	82.1	20.6	7.5
100 %	663'136	109.5	27.5	10.0
150 %	828'920	136.9	34.4	12.5
200 %	994'704	164.2	41.2	14.9
400 %	1'657'840	273.7	68.7	24.9
500%	1'989'408	328.5	82.5	29.9

Note: Own illustration. Created using Figma & Python.

To reach the 30-day target in Scenario B, UBS Group would have to triple its HQLA to around USD 907'000 million. For Scenario C, a sixfold increase to around USD 1'820'000 million would be required, which exceeds UBS Group's current balance sheet total of approx. USD 1'700'000 million and is therefore not feasible. The problem thus lies in the outflow rates, not in the HQLA volume. This result is in line with the argument of Hellwig (2010, pp. 40–41), who shows that capital increases alone cannot eliminate the systemic fragility of large banks as long as the underlying maturity transformation persists.

A bank-specific calibration of the parameters k and f for UBS would be desirable but would require granular daily outflow data that are not publicly available.

These findings immediately raise the question of how regulation has responded to the phenomenon of digital bank runs and whether the most recent reforms close the identified gap.

4.3.2. Regulatory Classification

The Federal Department of Finance (EFD) states in the TBTF report of April 2024 that the liquidity outflows at CS revealed a hitherto new dimension and that the TBTF framework must in future increasingly also take into account scenarios with extremely high and rapid liquidity outflows (EFD, 2024, p. 38). As measure 24, the reform provides for a critical review of the LCR outflow factors (EFD, 2024, p. 55). The tightened TBTF liquidity requirements for SIBs, which since 1 January 2024 prescribe a 90-day instead of a 30-day horizon, are a step in the right direction (EFD, 2024, pp. 54–55; LiqV, Art. 31c in conjunction with Art. 23 para. 1). The sensitivity analysis presented here shows, however, that even a tripling of HQLA in Scenario B would only just reach the 30-day target, but would miss the 90-day requirement. Instead, the reform relies on expanding the *lender-of-last-resort* potential of the SNB and on legally anchoring the *public liquidity backstop* (EFD, 2024, pp. 55–56). This architecture implicitly admits that a bank like UBS Group cannot survive a digital bank run on its own.

5. Conclusion

This paper has examined how long UBS Group would withstand a digital bank run that follows the outflow pattern of the Credit Suisse crisis of March 2023.

The starting point was the theoretical insight that maturity transformation makes banks structurally vulnerable to coordinated deposit withdrawals (Diamond & Dybvig, 1983). The *Liquidity Coverage Ratio* under Basel III addresses this risk through a 30-day liquidity buffer. The analysis in Chapter 2.3 has shown, however, that digitalisation fundamentally erodes the time assumptions underlying the model: real-time transaction systems, social media as a coordination mechanism, and the removal of psychological frictions compress outflow dynamics into a few days.

The empirical reconstruction of CS outflow rates in Chapter 3 has quantified this thesis: the average daily outflow rate in the critical week was 2.24 times the LCR standard. The stress test built on it in Chapter 4 yields three central findings. First, UBS Group comfortably meets the regulatory 30-day criterion in the normal stress case (Scenario A: 54.7 days). Second, with CS-analogous outflow rates the HQLA base would be exhausted after around 14 days (Scenario B), so the 30-day window could not even be used for half its length. Third, in the extreme scenario of a digital flash run only 5 days remain (Scenario C). The sensitivity analysis shows that the 30-day threshold is endangered from a k-factor of 1.4 onwards and that a solution via higher HQLA holdings would not be economically feasible.

The methodological limitations of the model presented in Chapter 4.2, in particular the linearity assumption, the limited transferability of bank-specific outflow rates, and the omission of systemic feedback effects, imply that the actual survival time could be shorter than stated.

For the regulatory debate, the results suggest that the calibration of the LCR outflow factors needs revision, as the Federal Council also demands in its TBTF report (EFD, 2024, p. 55). The current architecture, which relies on state backstops as the last line of defence, is a pragmatic but not a preventive solution. Further research could complement the linear scenarios used here with nonlinear models that endogenise feedback effects between outflows and market liquidity. Likewise,

a comparison of the survival time of other G-SIBs under identical stress parameters would be informative in order to examine whether the weakness identified here is a systemic or a bank-specific phenomenon.

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List of Aids Used

Tool	Use
Claude Sonnet 4.6	**Proofreading:** • "Give me feedback on this paper. Take into account the requirements of these documents. Give me concrete, short bullet points." **Paraphrasing:** • "Rewrite the given text to a scientific standard. Use these papers as a guide" • "Shorten the given text without changing the content." • "Rewrite the given text to improve readability." • "I find the following text far too long and unnecessarily complicated for the reader. Rewrite it without losing the points." **Debugging:** • "What is wrong with this code? " **Revision:** • "I am an examiner at HSG, correct the following paper according to these criteria and give me feedback, then assign a grade"
Visual Studio Code	The software was used as a code editor.
Jupyter Note-book	The software was used to structure and edit the code.
Pandas	The library was used to create dataframes in Python in a simplified way.
Endnote 21	For citations and the bibliography of those sources that were recognised.
Figma	Creation of all tables in this paper.
Google Scholar	I used the search engine to search for scientific sources.
GitHub	GitHub was used to provide the Python code and the Excel table to the instructor.

Appendix

The calculation models used for the stress tests in Chapter 4 as well as the underlying data are stored in a repository on GitHub. They can be found at:

https://github.com/WHA-VWL-113/Anhang-UBS-Liquidity-Stress-Test/releases/tag/v1.0

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